



Original Article

Machine Learning-Based Prediction of Local Recurrence in Uveal Melanoma After Ruthenium-106 Plaque Brachytherapy Using Ultrasound Images and Clinical Data

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Abstract

Aims: Uveal melanoma (UM) is the most common intraocular malignancy in adults, with local recurrence probability depending on treatment center strategies, radioisotopes, and follow-up period after conservative treatments such as plaque brachytherapy. This study is the first to assess the effectiveness of machine learning (ML) models in predicting UM local recurrence following Ruthenium-106 plaque brachytherapy by integrating demographic and clinical data.

Materials and methods: Data from 167 UM patients treated with Ru-106 plaque brachytherapy between 2011 and 2021 were analyzed in this retrospective, single-center study. The average follow-up duration for assessing local recurrence was approximately 93 months. Demographic and clinical variables were collected as features, with local recurrence status. Feature selection was achieved using recursive feature elimination (RFE), and principal component analysis (PCA) was applied for dimensionality reduction. Six classifiers—logistic regression, random forest (RF), support vector machine (SVM), gradient boosting, AdaBoost, and XGBoost—were trained and evaluated.

Results: The RF model demonstrated optimal performance, achieving an area under the curve (AUC) of 0.86, accuracy of 0.82, precision of 0.77, recall of 0.91, and an F1 score of 0.83. The final features, ranked by importance from SHapley Additive exPlanations (SHAP) analysis of the best-performing model, included the largest basal diameter (LBD), plaque size, thickness, apex dose, diabetes mellitus, and transpupillary thermotherapy.

Conclusion: ML models, particularly RF, effectively predict UM local recurrence using combined demographic and clinical data. These models can support personalized treatment strategies to improve patient outcomes. Future studies with larger, multicenter cohorts are needed to refine predictions further.

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Key words: Machine learning; recurrence; Ru-106 plaque brachytherapy; uveal melanoma

Introduction

Uveal melanoma (UM) is the most widespread form of intraocular malignancy in adults, originating in the uvea, with approximately 90% of cases occurring in the choroid, 7% in the ciliary body, and 3% in the iris [1–3]. Several

treatment options exist to preserve the eye and maintain vision [4,5], including plaque radiotherapy, proton therapy, gamma knife radiosurgery, photodynamic therapy, transpupillary thermotherapy (TTT), and surgical procedures [6,7]. Among these, plaque brachytherapy remains the predominant globe-conserving treatment for UM. Radioactive plaques, such as Iodine-125 and ruthenium-106, are commonly used, with ruthenium-106, a beta-emitter, being particularly important for treating UM [8–10].

Ultrasound images (A-mode, B-mode, and ultrasound biomicroscopy [UBM]) are widely regarded as the primary diagnostic modality for evaluating intraocular melanoma, allowing detailed assessment of tumour characteristics such as internal structure, shape, and size [11–14]. It also

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